

YuJia Liang

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EDUCATION

National Taiwan Ocean University

Keelung, Taiwan

B.S. in Computer Science and Engineering - GPA: 3.6/4.0

Sep. 2022 – Jun. 2026

Relevant Courses: Data Structures & Algorithms, Graph Theory Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning, Natural Language Processing, Generative AI, Object-Oriented Programming, Cyber Security, Linear Algebra, Discrete Mathematics, Probability Theory

EXPERIENCE

ITRI (Industrial Technology Research Institute)

Jun. 2025 – Aug. 2025

AI Image Recognition Intern, Electronics and Optoelectronics Research Laboratories

Hsinchu, Taiwan

- Optimized multi-task YOLO model for autonomous driving, improving inference speed while maintaining driving area accuracy and detection recall
- Implemented real-time image streaming and detection result transmission via Ethernet/ROS, processing 30 FPS with sub-50ms latency
- Applied neural network pruning techniques to YOLO models for edge deployment
- Supported research initiatives by reproducing and exploring state-of-the-art (SOTA) models to benchmark performance and identify potential improvements

RESEARCH & PROJECTS

RiffNode – AI Guitar Effects Studio

2026

SwiftUI, AVFoundation, Accelerate/vDSP

WWDC26 Swift Student Challenge

- Selected as an Apple WWDC26 Swift Student Challenge Winner** – one of only 8 winners in Taiwan and 350 worldwide – for a professional guitar effects app built entirely with native Apple frameworks (no third-party APIs), compressing to under 2 MB; featured in national press ([UDN](#), [CNA](#))
- Built a dual-AI tone engine: Apple Foundation Models generates structured DSP parameters on-device via @Generable while Gemini 2.0 Flash enriches explanations, staying fully functional offline
- Engineered a low-latency real-time audio pipeline with AVAudioEngine driving 11 professional effects (compressor, 10-band EQ, overdrive, distortion, reverb, delay, modulation) in a drag-and-drop signal chain with live reconnect
- Implemented hands-free control using Vision Framework face-landmark detection (nod, tilt, mouth-open gestures) plus real-time Accelerate/vDSP FFT analysis, autocorrelation pitch detection, tuner, metronome, and chord-aware tone suggestions

Volleyball Performance Analysis System – Senior Capstone

Nov. 2024 – Dec. 2025

Python, PyTorch, YOLO, OpenCV, React, TypeScript, FastAPI, Celery, Redis, Docker, pytest Advisor: Prof. Pei-Yi Ding

- Developed comprehensive volleyball analytics system achieving 94.49% mAP@0.5 for action recognition (5 classes: serve, spike, block, receive, set) and 79.5% ball tracking accuracy
- Architected distributed microservices with FastAPI backend, Celery task queue, and Redis message broker; implemented WebSocket endpoints for real-time analysis progress tracking
- Built multi-model deep learning pipeline: VballNet (U-Net ONNX) for ball tracking at 200+ FPS, YOLOv11m for action detection, YOLOv8+Norfair for player tracking with 87.6% ID consistency
- Developed React/TypeScript frontend with Video.js player, Recharts analytics dashboard, interactive timeline with frame-level seeking, and real-time bounding box overlays
- Implemented trajectory filtering pipeline with velocity-based outlier removal, Gaussian smoothing, and polynomial interpolation; action consolidation algorithm reducing fragmented events by 45%
- Containerized application with Docker Compose orchestrating 5 services; configured GitHub Actions CI/CD with flake8 linting and automated builds
- Wrote comprehensive test suite using pytest with async support, achieving coverage across API endpoints, database operations, and ML inference pipelines

Ghote Notes – AI Note-Taking Application

Nov. 2025 – Present

Tauri 2.0, Rust, React 19, TypeScript, Lexical, D3.js, Supabase, PartyKit, Yjs, Vitest, Playwright Individual Project

- Built privacy-focused macOS note-taking app using Tauri 2.0 (Rust) with React 19 frontend, featuring vault-based storage with Obsidian-compatible markdown files
- Implemented rich text editing with Facebook's Lexical framework, supporting slash commands, KaTeX math equations, Mermaid diagrams, and bidirectional markdown sync

- Developed interactive knowledge graph using D3.js with wiki-style `[[link]]` parsing, enabling visual exploration of note relationships
- Built local RAG system using Transformers.js with all-MiniLM-L12-v2 embeddings (384-dim) for semantic search, featuring smart chunking and cosine similarity ranking
- Implemented real-time collaboration via PartyKit WebSocket server with Yjs CRDT for conflict-free concurrent editing and cursor presence
- Integrated Supabase cloud sync with PostgreSQL RPC functions and multi-device vault synchronization
- Achieved 88.75% test coverage with 521 unit tests using Vitest; built comprehensive Tauri API mocks for file system, dialogs, and path utilities

SyncUp – Social Calendar App

2025

Flutter, Dart, Firebase, Google Maps API, Google Gemini API

Individual Project

- Developed social calendar app with real-time sharing and friend management using Flutter and Firebase
- Implemented location-based event suggestions with Google Maps integration and Firebase Authentication
- Built AI-powered chatbot connected with Google Gemini to intelligently generate schedules and suggest meetup opportunities based on user preferences

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript/TypeScript, Rust, Swift, Dart, SQL
ML/AI: PyTorch, TensorFlow, OpenCV, ONNX Runtime, scikit-learn, NumPy, Pandas, CreateML, Foundation Models, Vision Framework, Transformers.js, Norfair
Frameworks: React, Flutter, FastAPI, Node.js, SwiftUI, AVFoundation, Accelerate/vDSP, Tauri, Lexical, Celery, Video.js, Recharts, D3.js
DevOps & Tools: Git, Docker, GitHub Actions, pytest, Playwright, Vitest, Redis, PostgreSQL, SQLite, Supabase, Firebase
Languages: Mandarin (Native), English (Fluent, TOEFL 110, TOEIC 920), Japanese (JLPT N3)

LEADERSHIP & ACTIVITIES

Volleyball Department Team Captain	2022 – present
	<i>Keelung, Taiwan</i>
NTU Photography Club (NTUPhoto)	2025 – 2026
<i>Member & Darkroom Officer</i>	<i>Taipei, Taiwan</i>
NTUFC Kronos Quantitative Reading Club	Sep. – Dec. 2025
Hackathon Participant	2022 – present